

Assignment 4 Report

Rahul Vimalkanth
EE23B061

October 2024

1 Introduction

The objective of this project is to visualize key usage with a heatmap and calculate the total finger travel distance across the keyboard. This analysis aids in understanding typing efficiency, informing decisions about keyboard layouts, and potentially leading to optimized typing experiences.

2 Approach

The program follows these main steps:

1. **Key Mapping and Layout:** Keyboard layouts, such as QWERTY and Dvorak, are defined, with each key mapped to specific positions using 2D coordinates. Both layouts can be passed as input arguments.
2. **Text Analysis:** The input text is analyzed, counting the frequency of each key press, including special keys like 'Shift' for uppercase letters and symbols.
3. **Distance Calculation:** The Euclidean distance between key positions is calculated for each character.
4. **Heatmap Generation:** A heatmap is generated using the key press frequencies, with frequently used keys appearing in hotter colors (red) and infrequent keys in cooler colors (blue/purple).

3 Assumptions

- The default layout is QWERTY, but the program supports alternative layouts like Dvorak.
- The space bar is considered a home row key, with a travel distance of zero. It is not displayed on the heatmap for better visualization of other keys.
- The left side of the keyboard is mapped to the right 'Shift' key, and the right side to the left 'Shift' key for efficiency.
- The distance calculation considers only one-way movement to each key, excluding the return trip to the home row.
- Shifted characters for non-alphabet symbols are displayed within the corresponding key's patch at the top right corner.
- Extra keys like 'Fn', 'Win', 'Alt', and 'Ctrl' are included in the layout but not used in distance calculations.

4 Implementation Details

4.1 Key Mapping

Each key has a specific coordinate. For example, the key 'f' is at position (4.75, 2), and 'y' is at (6.5, 3). The coordinates reflect the keyboard layout, with the x-axis increasing from left to right and the y-axis from bottom to top.

4.2 Text Analysis and Finger Travel Distance

For each character in the input text, the corresponding key or combination of keys is identified. The Euclidean distance is computed using:

$$\text{distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

where (x_1, y_1) and (x_2, y_2) are the coordinates of the starting and target keys, respectively. The total distance is summed over the entire text, with special handling for shifted characters.

4.3 Heatmap Visualization

A heatmap is generated from key press frequencies. Higher frequencies are displayed in warmer colors (red), while lower frequencies appear in cooler colors (blue/purple). Non-alphabet symbols that require the ‘Shift’ key show their shifted characters at the top right of the keys.

Figures 1 and 2 show the heatmaps generated for the input text using the QWERTY and Dvorak layouts, respectively.

5 Findings

For the input text *“One of the most important things in life is to be happy. It’s all that matters!”*, the total finger travel distances were calculated as follows:

- **QWERTY:** 61.93 units
- **Dvorak:** 36.81 units

5.1 Key Usage Insights

The heatmap analysis revealed that the most frequently used keys were:

- **Space Bar:** Commonly used for word separation.
- **Vowels:** Keys such as ‘a’, ‘e’, ‘i’, ‘o’, and ‘u’ had higher usage due to their prevalence in English.
- **Common Consonants:** Keys like ‘t’ and ‘s’ were heavily pressed.

5.2 Comparative Layout Analysis

The Dvorak layout resulted in a significantly lower total finger travel distance compared to QWERTY. This efficiency stems from the placement of frequently used letters in the home row, which minimizes finger movement and potentially enhances typing speed and comfort.

6 Conclusion

This program provides insights into keyboard usage patterns and finger travel distances. The analysis indicates that the Dvorak layout reduces finger travel distance compared to QWERTY, suggesting it may be more efficient for frequent typists.

Future work could explore customized layouts optimized for individual users’ typing patterns and implement adaptive feedback mechanisms to dynamically adjust layouts based on actual usage.



Figure 1: Heatmap of keyboard usage for the QWERTY layout.

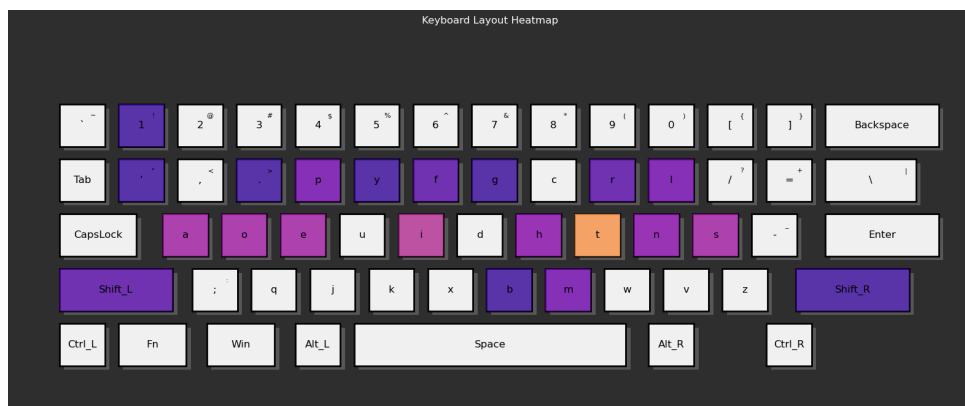


Figure 2: Heatmap of keyboard usage for the Dvorak layout.